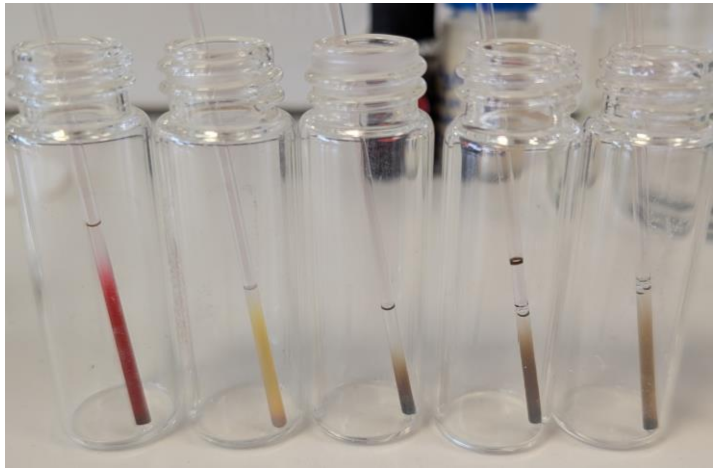
Problem :

Measure the concentration of nanoparticles.

For the gold it is possible with the UV-vis spectroscopy at 400 nm, but not for the other elements.

Measure by SAXS :

- Use calibrated capillaries (measure at 1 cm from the bottom)
- Use the same time measurements for all the samples.
- Measurr Li34, air, an empty capillary and the glassy carbon in the same conditions (be careful, use the fixed offset)
- Fill properly all the date in the SAXS software (thickness, time, comments…). Do not rename any file.
- Use freshly prepared capillaries if the particles tend to fall to the bottom.

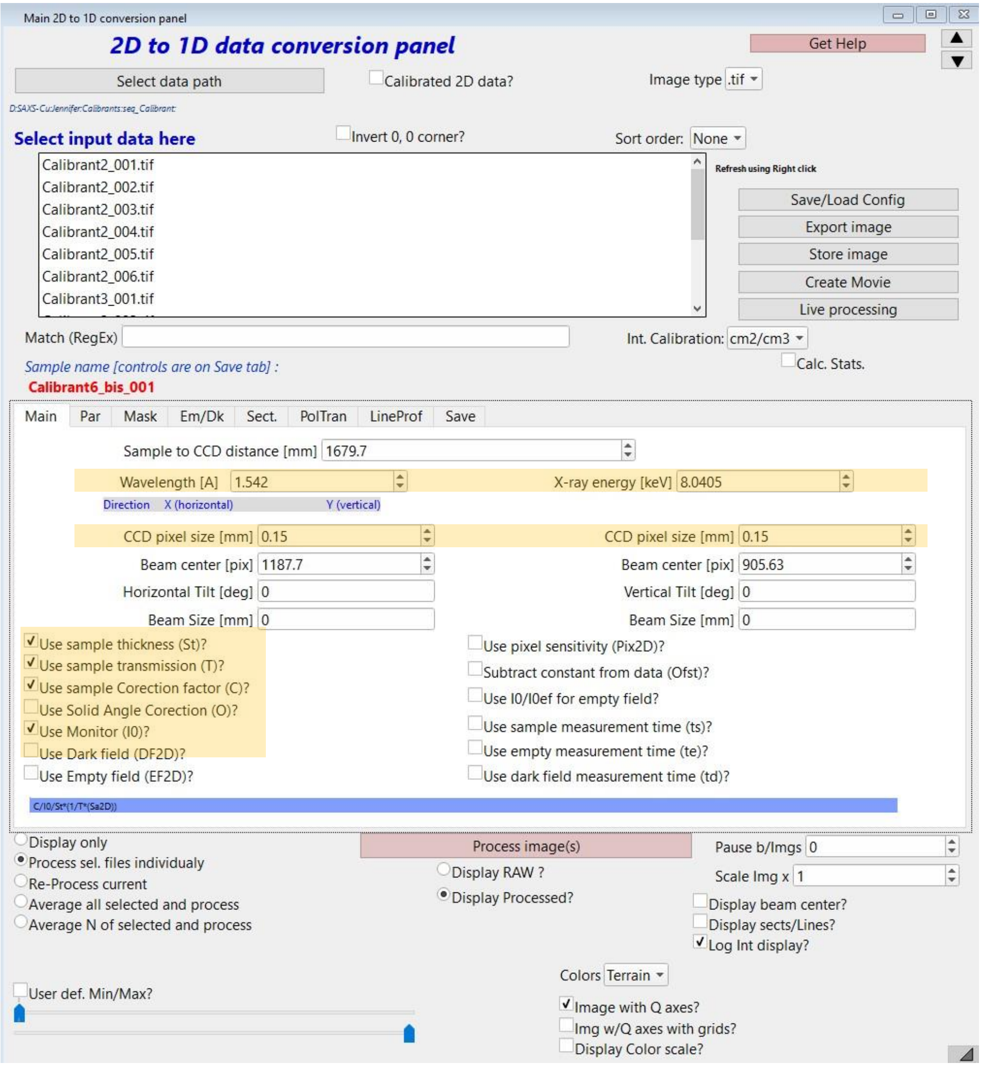


Au Ag Pt Pd Rh

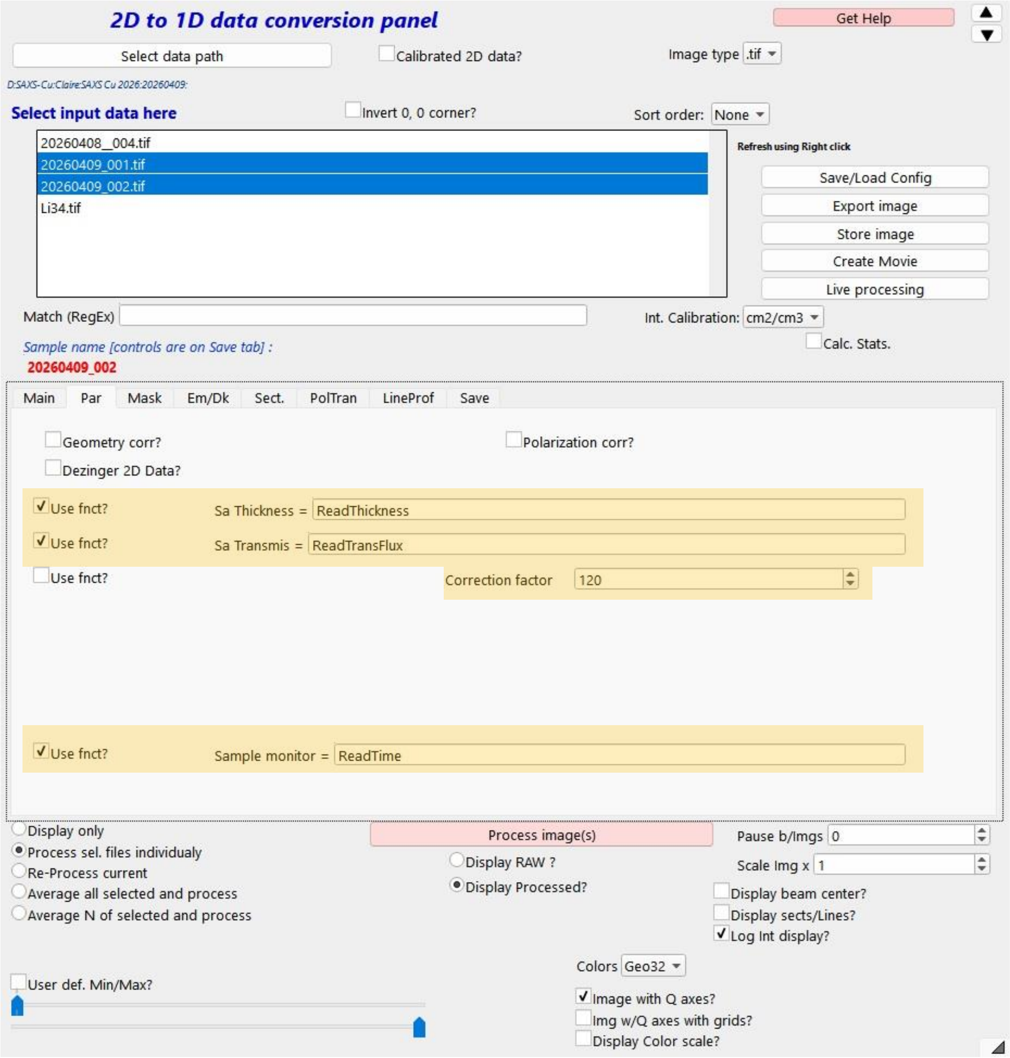
Analyse :

1) In Igor, use the usual data treatment (mask, calibration beam center Li34 etc).

2) Tick the good elements in the Main tab and use the good values. **Be careful**, these ones are for SAXS Cu, not SOLEIL !

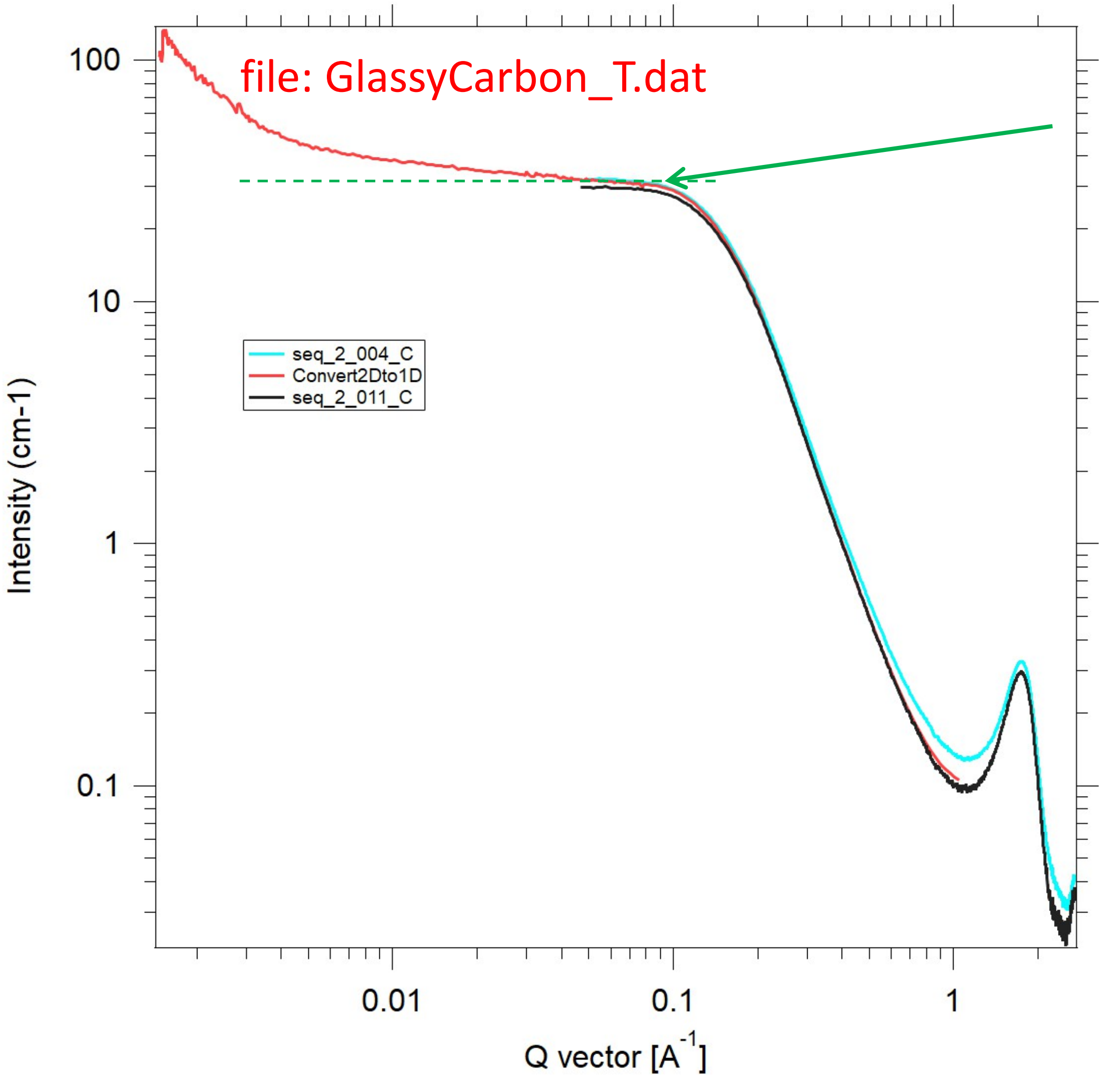


3) Par tab: use the functions and start with a correction factor of 120 (for SAXS Cu).



Glassy carbon

Absolute intensity in cm^{-1}



Plateau at 30 cm^{-1}

Calibration sample

l_{GC} = 1.16 mm

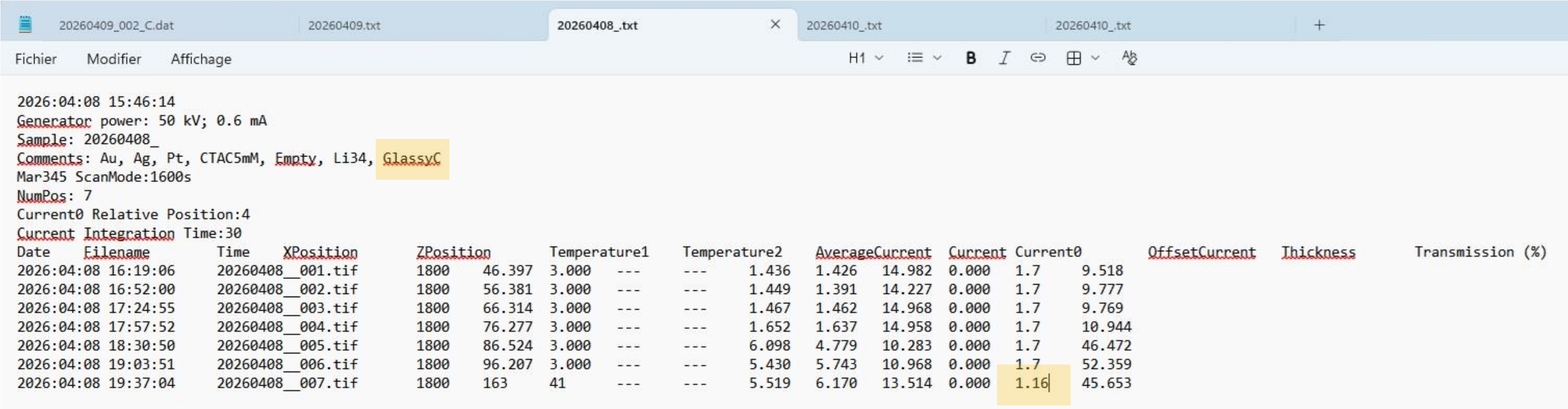
$correction\ factor$: C=120

in Igor Nika data reduction

for Cu-SAXS instrument

Igor Nika data reduction for Cu-SAXS instrument takes into account the thickness !

4) In the log file, modify the thickness of the glassy carbon if it’s not already done: 1,16 mm

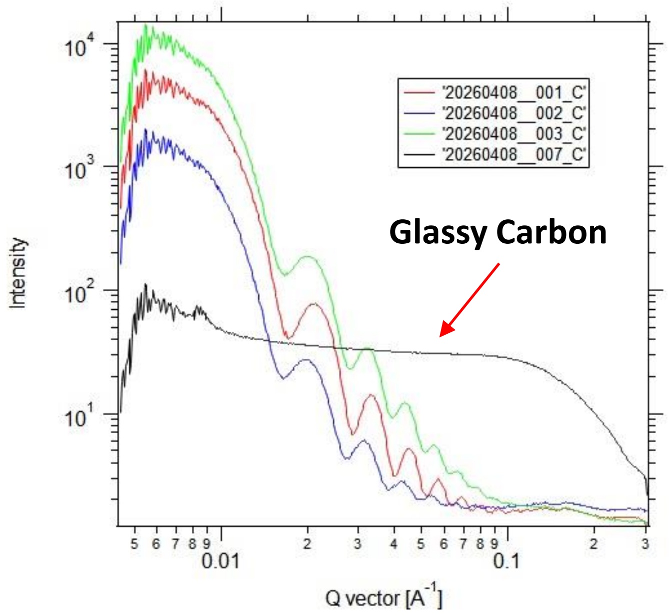


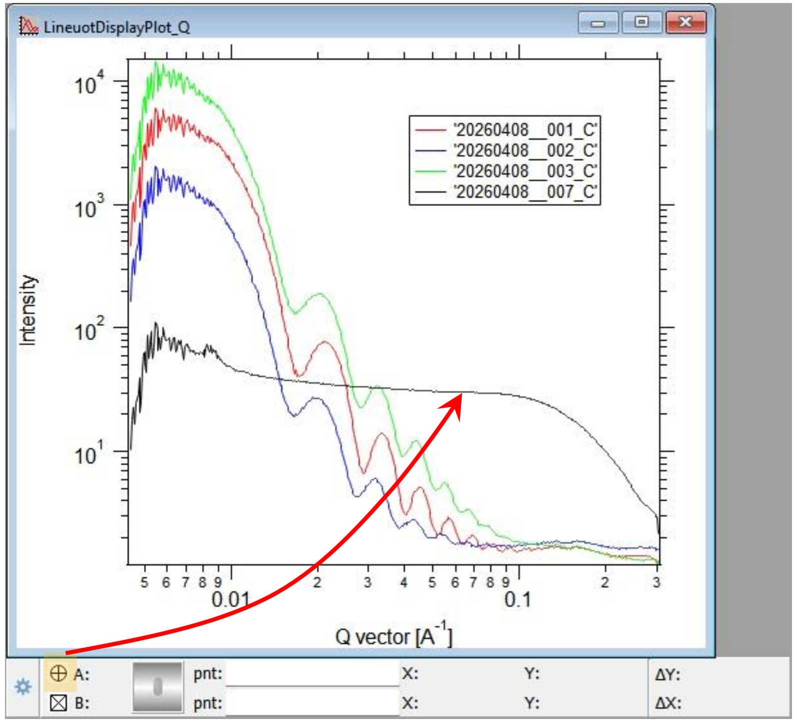
5) Laund a « process images »

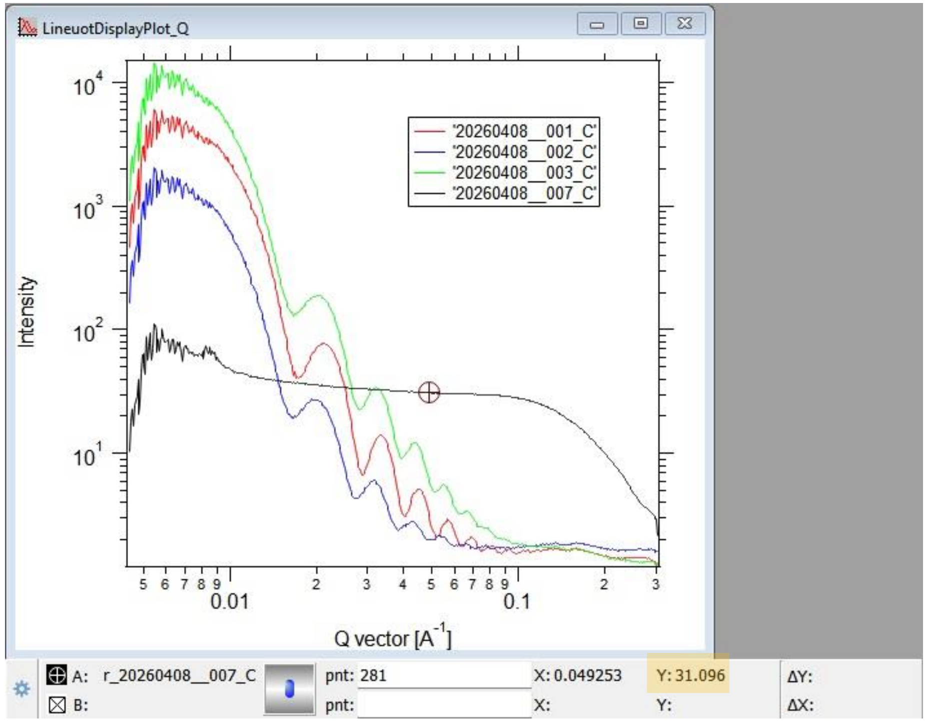
Optimisation of the correction factor (if needed)

The plateau of the Glassy Carbon must be around I = 30 (at 0,6 A^{-1}).
To read it properly in Igor, click on the graph window and predd CTRL+I.
A sub-window appears.
Take the cursor and place it on the plateau around at 0,06 A^{-1}.
Leave the cursor and reand the value of I.

If I is far from 30, modify the correction factor (do a cross product) et laund again a « process images ».





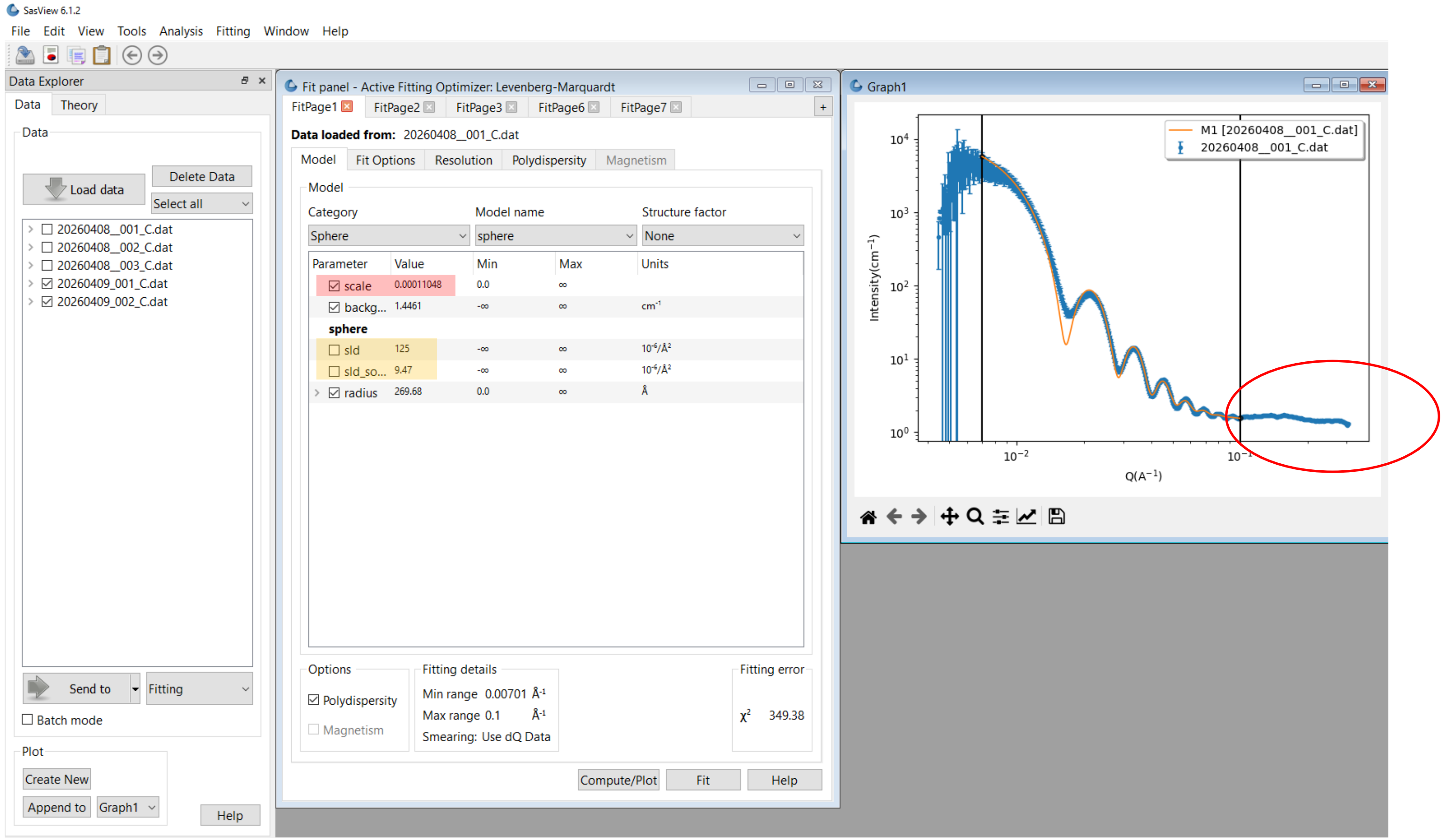


Analyse in SASview

Open the.dat files generates in Igor with SASview and send them to fitting without any additionnal operation.

It’s important to entend the SLD of the solvent and nanoparticles. We can find them in Tools → SLD Calculator (we need to know the density of the materials).

The value we are looking for is the « scale ». It corresponds to the value of the plateau for q = 0.



Il faudra implémenter la soustraction du empty (mesure dans l’air/vide) et du dark

The « scale » corresponds to the volume fraction of the nanoparticles in the sample.
Here I noted the scale value using a polydispersity or not, the fits were not perfect in both cases.
We need the density and molar weight of the materials.

	Au	Ag	Pt	Pd	Rh
d (g/cm3)	19,30	10,49	21,45	12,02	12,41
M (g/mol)	196,96	107,89	195,08	106,42	102,90
SLD (10^6 A^{-2})	125,00	77,90	136,00	88,50	92,30
Scale without PD	0,00009848	0,00008268	0,00017710	0,00007595	0,00004559
Scale with PD	0,00011048	0,00009444	0,00022793	0,00008806	0,00005608
Volume fraction (vol%)	0,01	0,01	0,02	0,01	0,00
Mass fraction (wt%)	0,19	0,09	0,38	0,09	0,06
[M] (mol/L)	0,01	0,01	0,02	0,01	0,01
[M] (mM)	9,65	8,04	19,47	8,58	5,50
Concentrations given by Sergio (mM)	9,80	17,62	11,02	12,36	?
Without PD	9,65	8,04	19,47	8,58	5,50
With PD	10,83	9,18	25,06	9,95	6,76

*100

*d

*10/M

*1000

For gold, the concentration measured with UV-vis fits well with the one measured by SAXS.

φ_i (wt%) $= 100 \dfrac{m_i}{m_{TOT}} \iff m_i = \varphi_i(\text{wt\%}) \times \dfrac{m_{TOT}}{100}$

Si je fixe $m_{TOT} = 1000$ g $\iff$ 1L on a donc

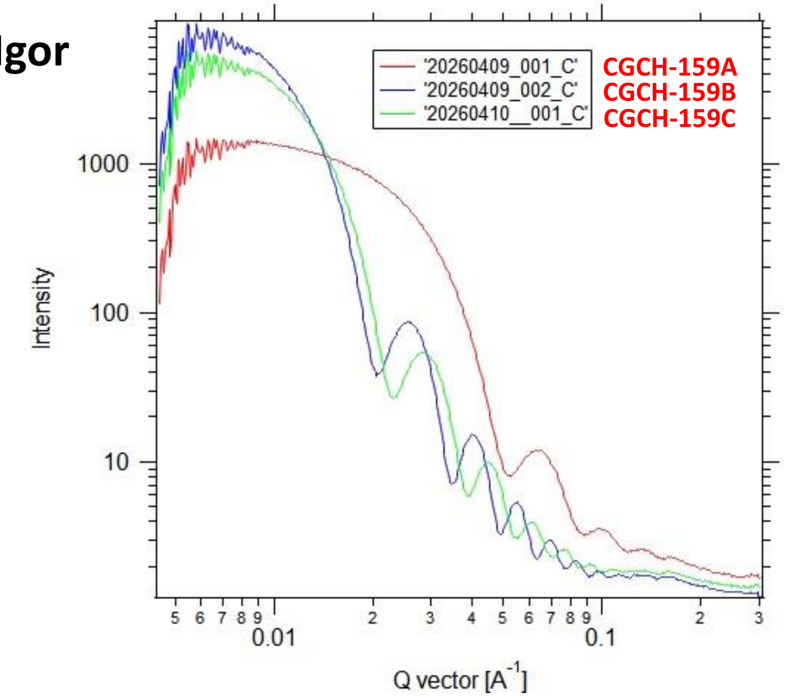
la concentration massique $Cm_i = 10\, \varphi_i(\text{wt\%})$ (en g/L)

Et donc la concentration molaire $[M_i] = \dfrac{Cm_i}{M} = \dfrac{10\,\varphi_i(\text{wt\%})}{M}$ (en mol/L)

My M is the molar weight in g/mol !

Gold spheres of different sizes

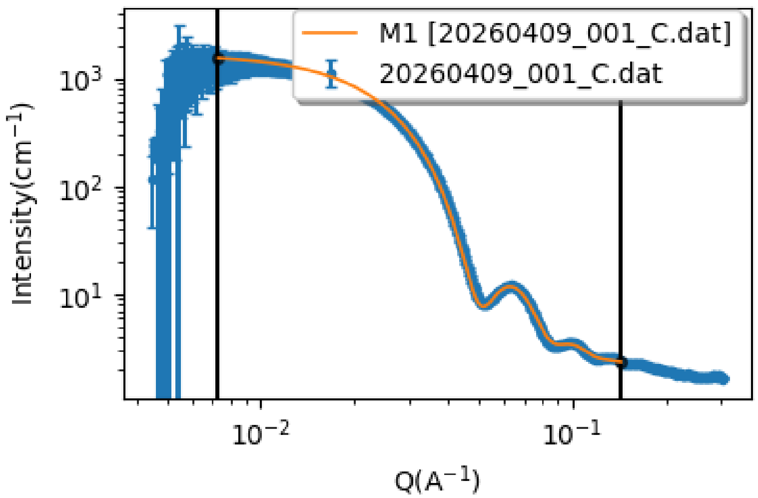
Traitement Igor



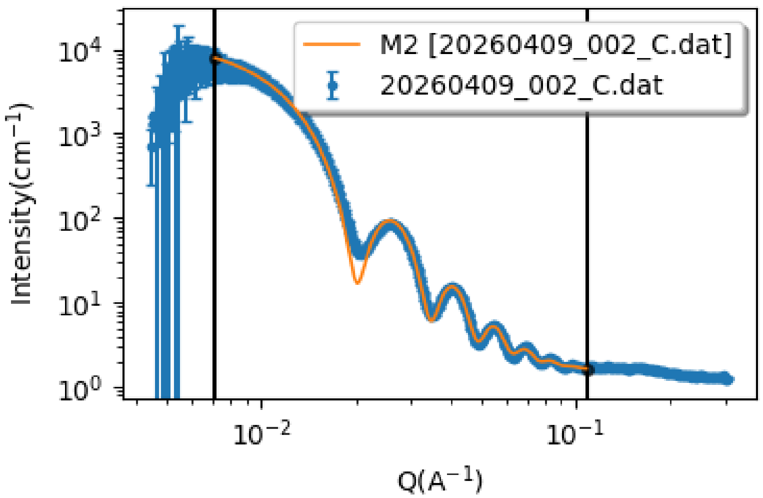
	CGCH-159A	CGCH-159B	CGCH-159C
d (g/cm3)	19,3	19,3	19,3
M (g/mol)	196,96	196,96	196,96
SLD (10^6 A-2)	125	125	125
Radius (A)	86,159	221,79	197,8
Diameter (nm)	17,2	44,4	39,6
PD (%)	10,4	5,2	6,7
Scale without PD	0,00035297	0,00018622	0,00016093
Scale with PD	0,00041759	0,00021437	0,00018876
Volume fraction (vol%)	0,041759	0,021437	0,018876
Mass fraction (wt%)	0,8059487	0,4137341	0,3643068
[M] (mol/L)	0,04091941	0,021005996	0,018496487
[M] (mM)	40,92	21,01	18,50
Concentrations given by UV (mM)	29,31	18,30	15,44
Without PD	34,59	18,25	15,77
With PD	40,92	21,01	18,50

Fits SASview

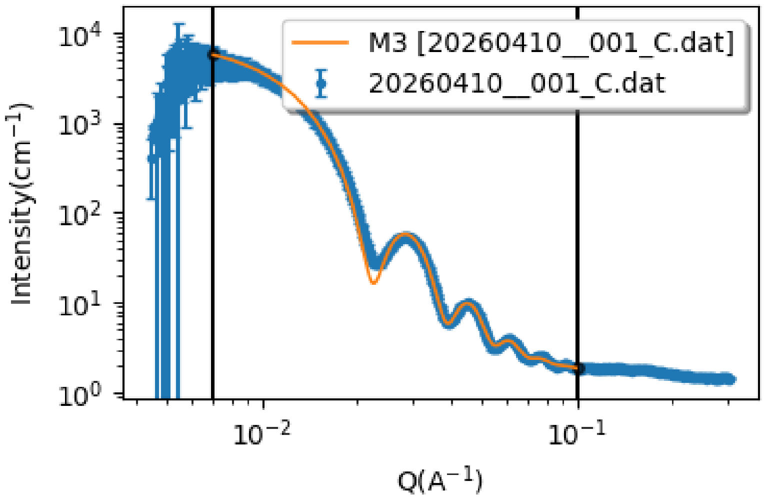
CGCH-159A



CGCH-159B



CGCH-159C



GC carbon calibration for SWING data

$$I_{GC}\,(cm^{-1}) = K\,\frac{I_{Swing\_GC} - I_{Swing\_air}}{l_{GC}\,(mm)}$$

In Foxtrot macro, use K=0.036 for the calibration constant (instead of K=1)

GC calibration: K=0.036

Superimpose GC (cm-1) with the calibration curve by Jan Ilavsky (file: GlassyCarbon_T.dat).
Calibration constant K should be close to 0.036.
l_{GC} = 1.16 mm

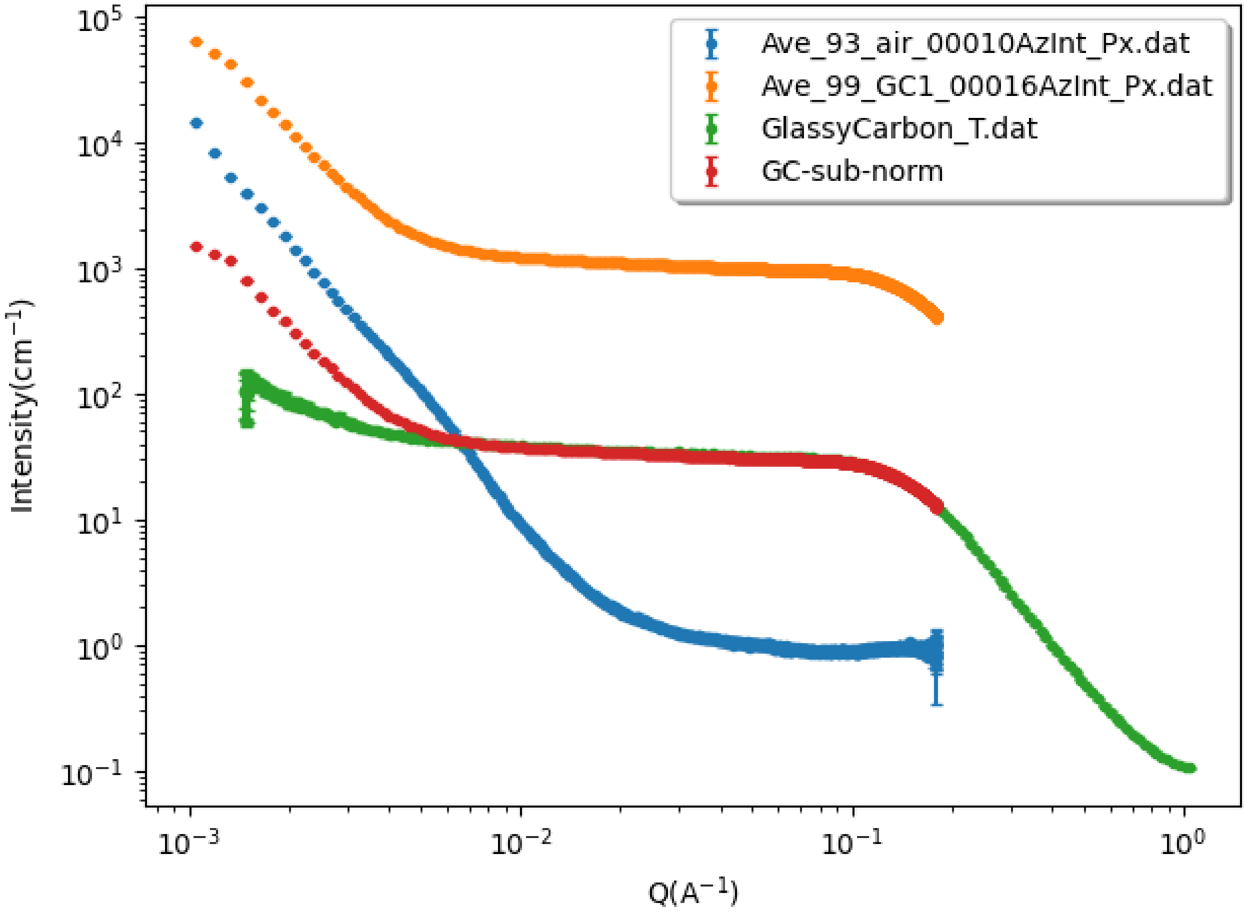
$$I_{cap\_norm\_sub}\,(cm^{-1}) = K\,\frac{I_{Swing\_cap} - I_{Swing\_cap\_solvent}}{l_{cap}\,(mm)}$$

Calibration of intensity for a sample in a glass capillary:

Take the same thickness l_{cap} (in mm) for the capillary filled with solvent

Thickness is not taken into account in Foxtrot data reduction

SWING data April 26: GC calibration



l_{GC} = 1.16 mm

GC calibration: K=0.036

$$I_{GC}\,(cm^{-1}) = K\,\frac{I_{Swing\_GC} - I_{Swing\_air}}{l_{GC}\,(mm)}$$

In SASView GUI, go to tools/data operation:

- Subtract air signal:

-> Compute and save $I_{Swing\_GC} - I_{Swing\_air}$

- Divide by thickness l_{GC} = 1.16 mm:

-> Compute and save $\frac{I_{Swing\_GC} - I_{Swing\_air}}{l_{GC}\,(mm)}$

- Multiply by calibration constant **K=0.036**
- -> Compute and save $I_{GC}\,(cm^{-1}) = K\,\frac{I_{Swing\_GC} - I_{Swing\_air}}{l_{GC}\,(mm)}$

- Verify that $I_{GC}\,(cm^{-1})$ superimposes with the calibration curve by Jan Ilavsky (file: GlassyCarbon_T.dat)

For all SWING data

In Foxtrot, use K=0.036 for the calibration constant (instead of K=1) and run again the macro

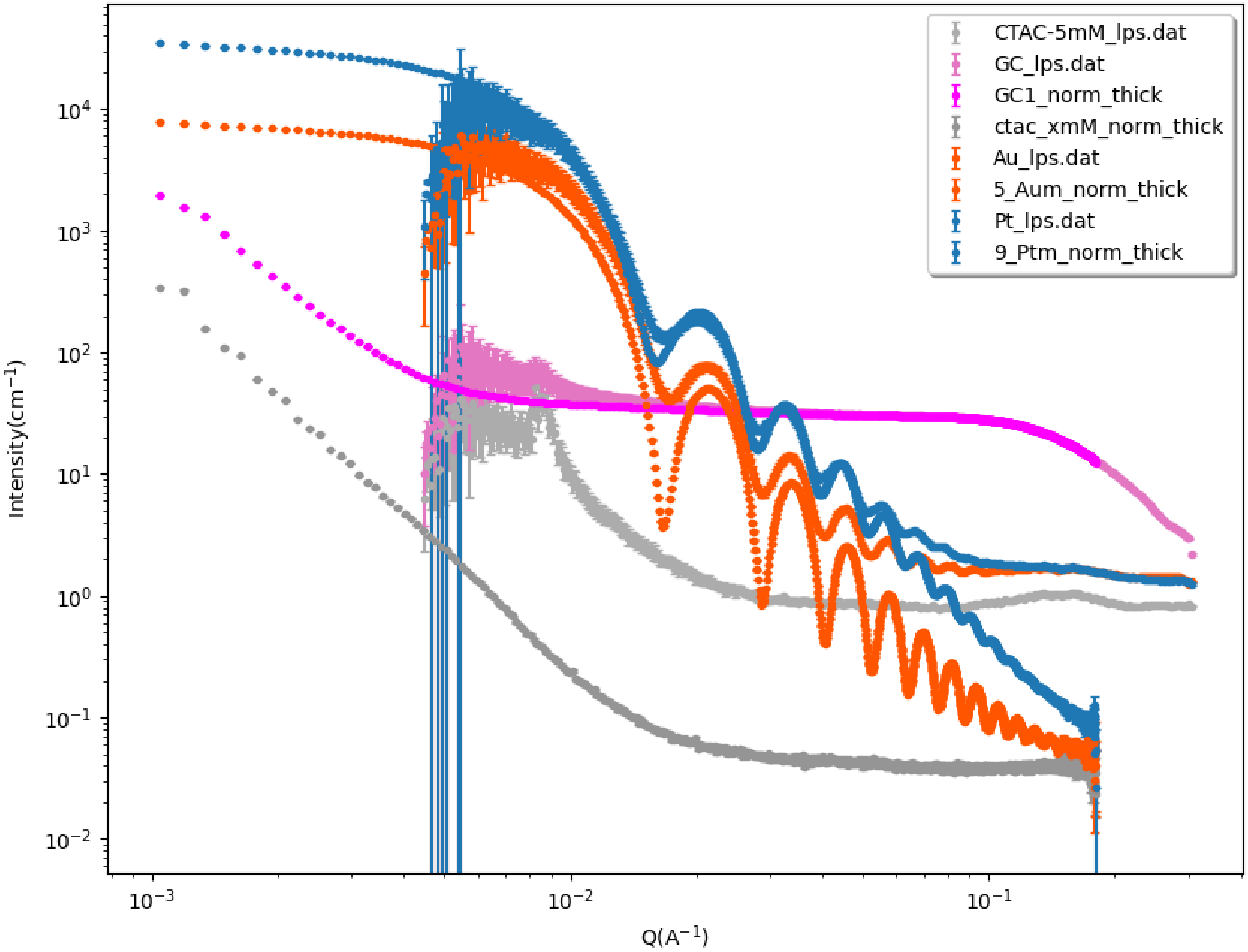
$$I_{cap\_norm\_sub}\,(cm^{-1}) = K\,\frac{I_{Swing\_cap} - I_{Swing\_cap\_solvent}}{l_{cap}\,(mm)} = K\,\frac{I_{Swing\_cap}}{l_{cap}\,(mm)} - K\,\frac{I_{Swing\_cap\_solvent}}{l_{cap}\,(mm)}$$

$$I_{cap\_norm}\,(cm^{-1}) = K\,\frac{I_{Swing\_cap}}{l_{cap}\,(mm)}$$

Calibration of intensity for a sample in a glass capillary:

** Take the same thickness l_{cap} (in mm) for the capillary filled with solvent

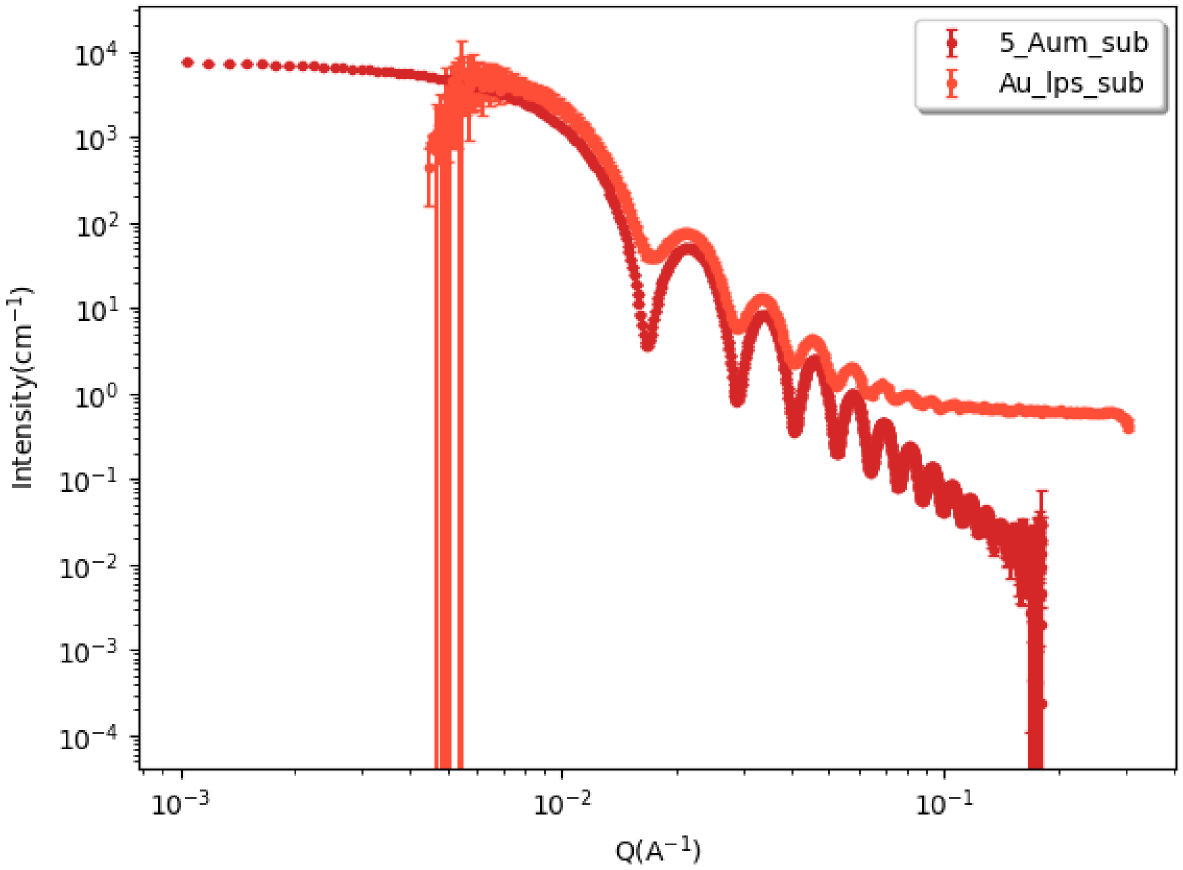
LPS and SWING data: Sergio's mother solutions for Au and Pt

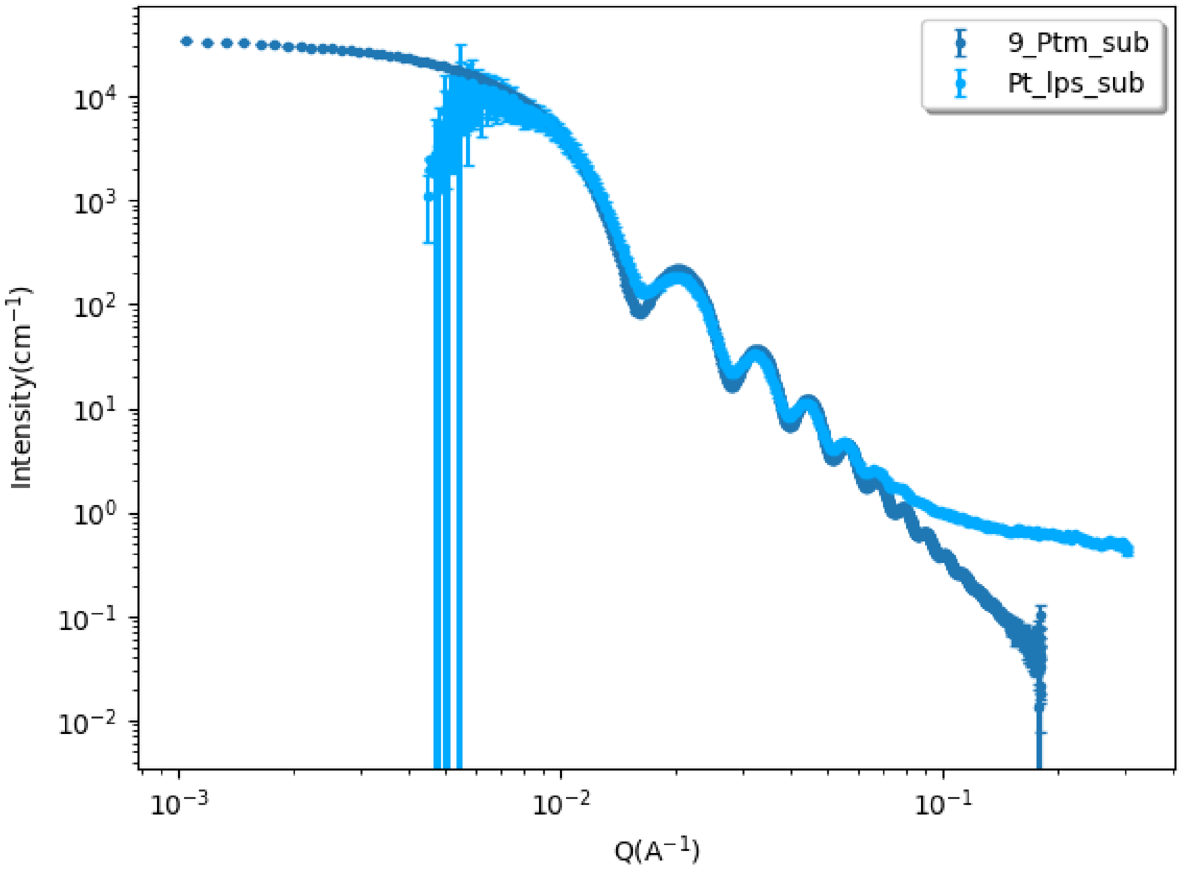


l_{GC} = 1.16 mm

l_{cap} = 1.7 mm

LPS and SWING data: Sergio's mother solutions for Au and Pt





with solvent subtraction

Absolute Intensity

What we measure during one acquisition == Number of photons in a pixel

$$N_{pixel}(\vec{q}) = \eta\ (\phi_0\ A\ T)\ \ t_{acq}\ \Delta\Omega\ \ l\ \ I(\vec{q}) + \mathrm{BG}$$

$[L^{-1}]$ $\quad I(\vec{q}) = \dfrac{1}{V_{tot}}\langle|A(\vec{q})|^2\rangle_t$

$[L]$ $\quad A(\vec{q}) = \displaystyle\int_{V_{tot}} \rho(\vec{r})e^{-i\vec{q}\cdot\vec{r}}d\vec{r}$

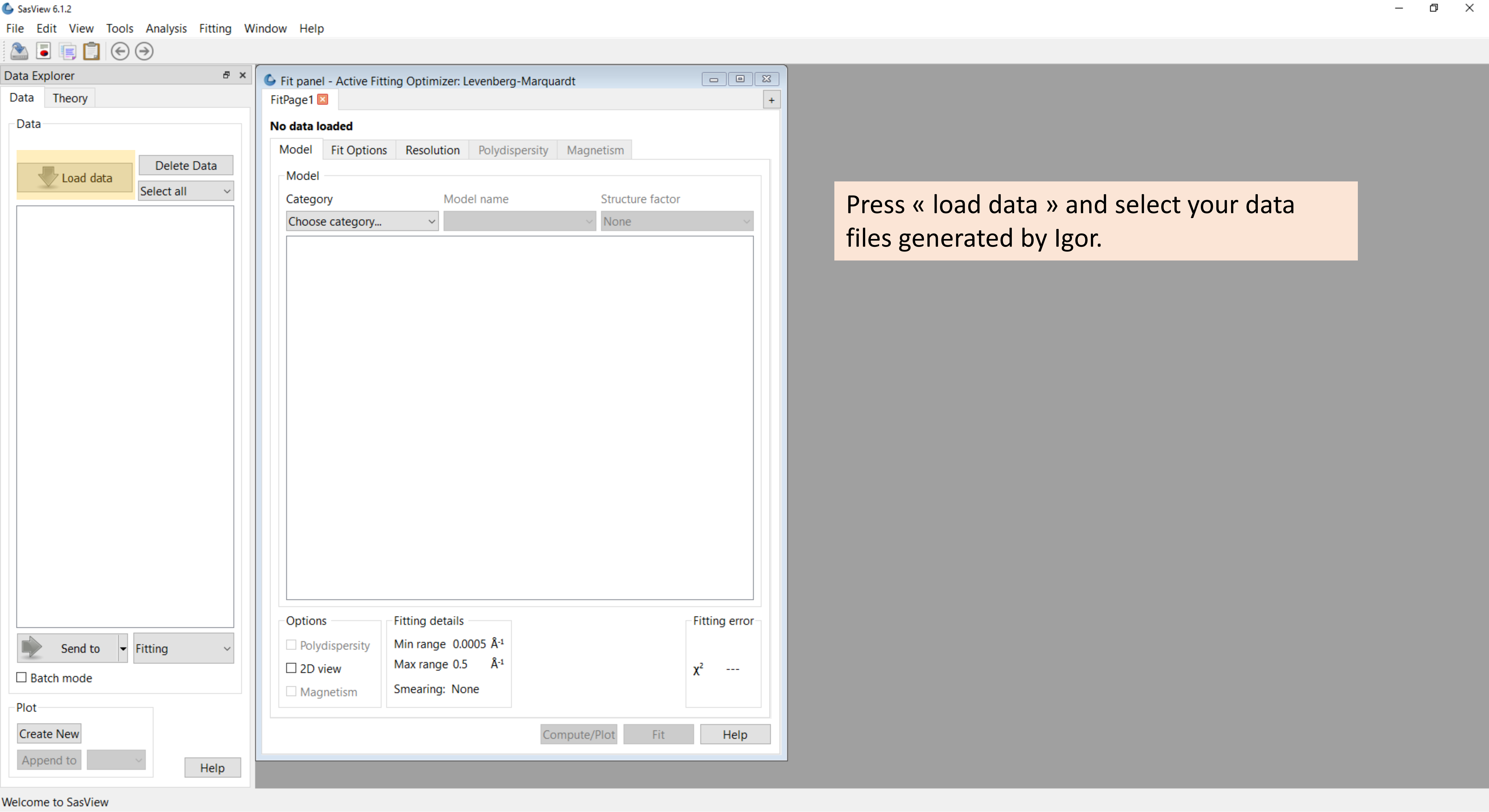
$[L^{-2}]$ $\quad \rho(\vec{r}) = r_e\ \rho_e(\vec{r})$: SLD

$[L^{-3}]$ $\quad \rho_e(\vec{r})$: electron density

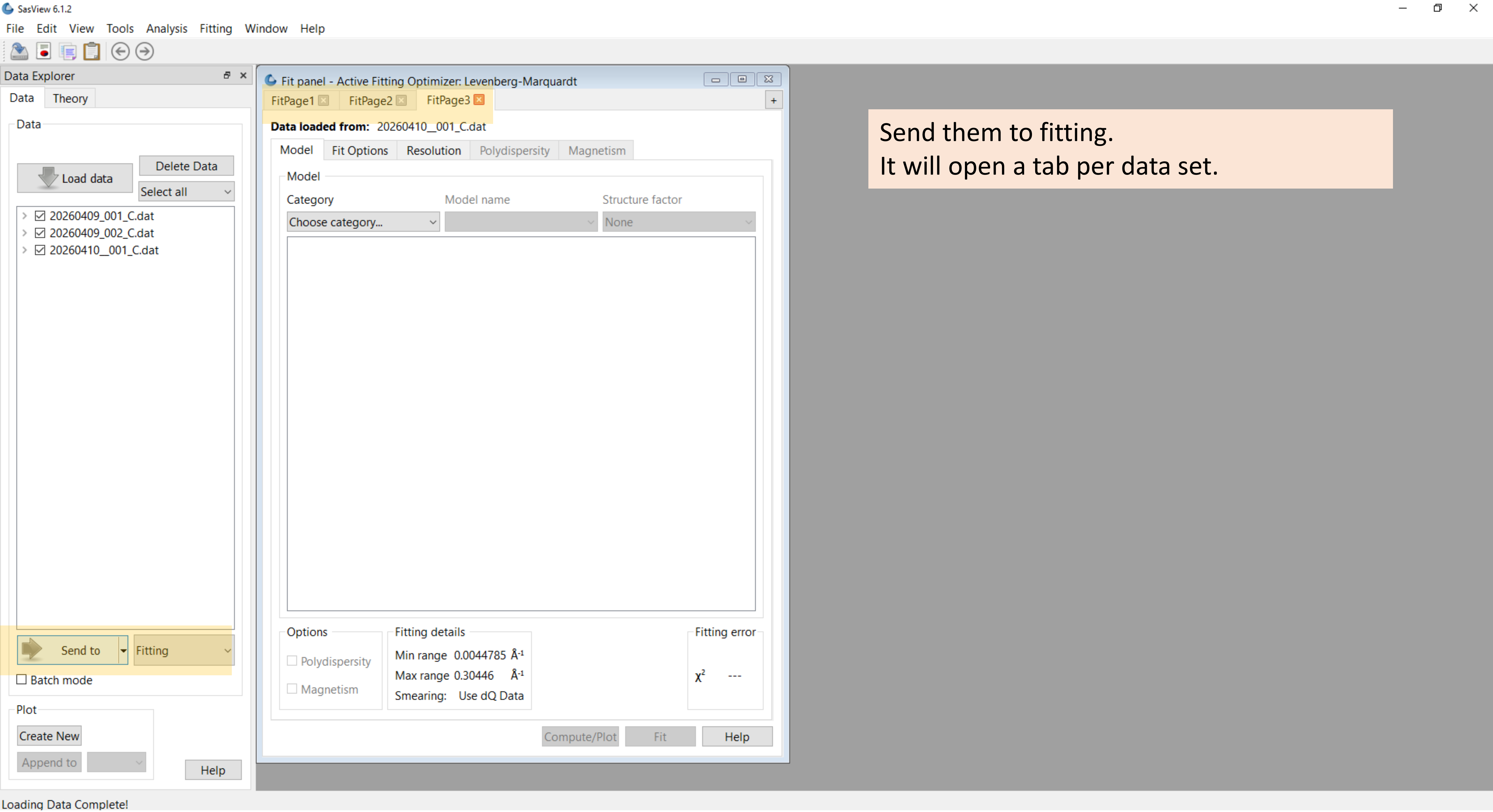
r_e = 2.81794×10^{-15} m
Classical radius of electron

- η: detector efficency
- ϕ_0: incident flux == photons number per unit time and surface area
- A : section of the X-ray beam on the sample
- T : sample transmission
- $(\phi_0\ A\ T)$: integrated transmitted flux
- t_{acq} : acquisition time
- l : sample thickness
- $A\ l = V_{tot}$: sample volume inside the X-ray beam
- Δ: pixel size; r : distance between sample and pixel
- $\Delta\Omega = \left(\dfrac{\Delta}{r}\right)^2$: solid angle for one pixel

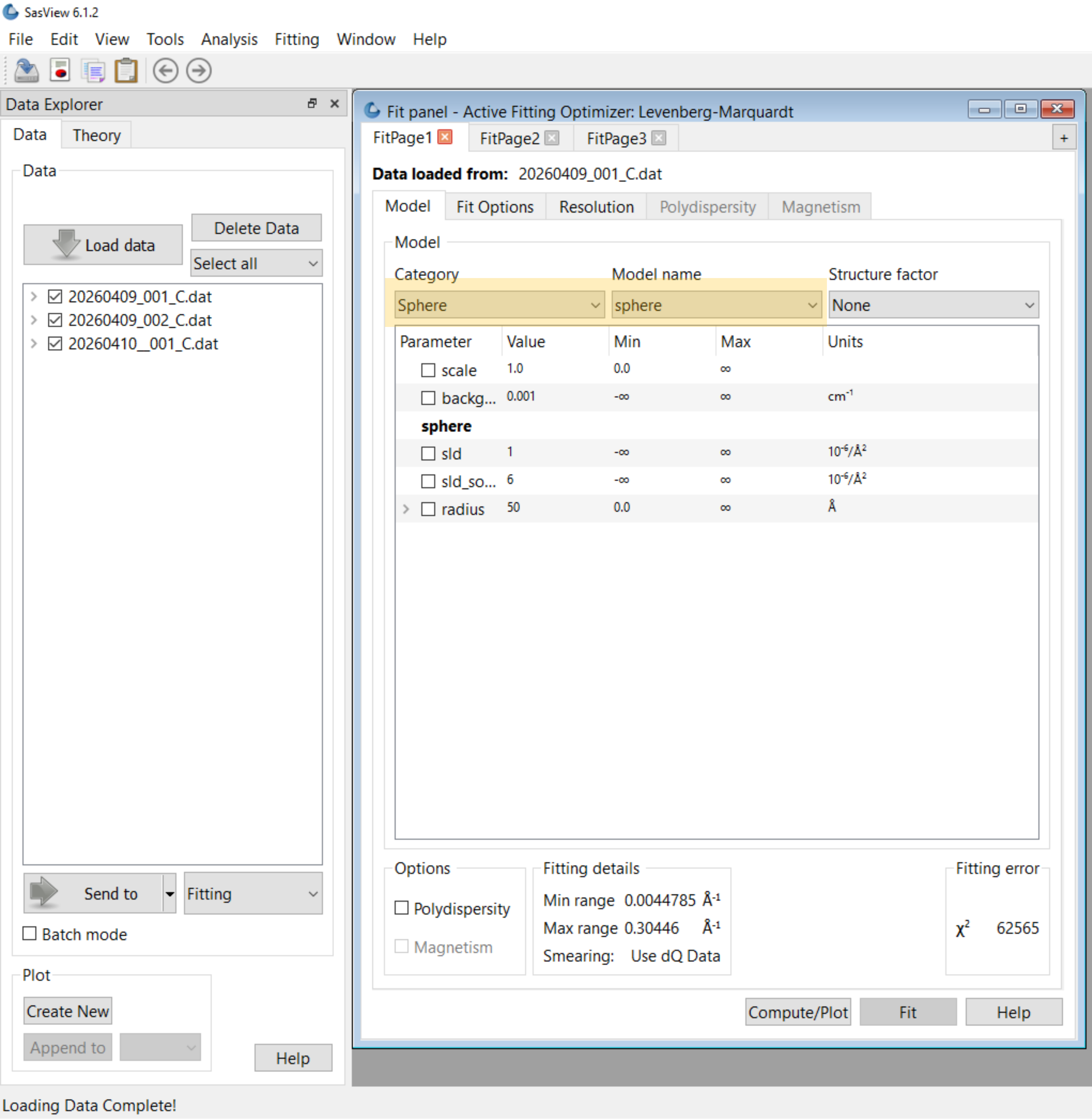
Tuto SASview



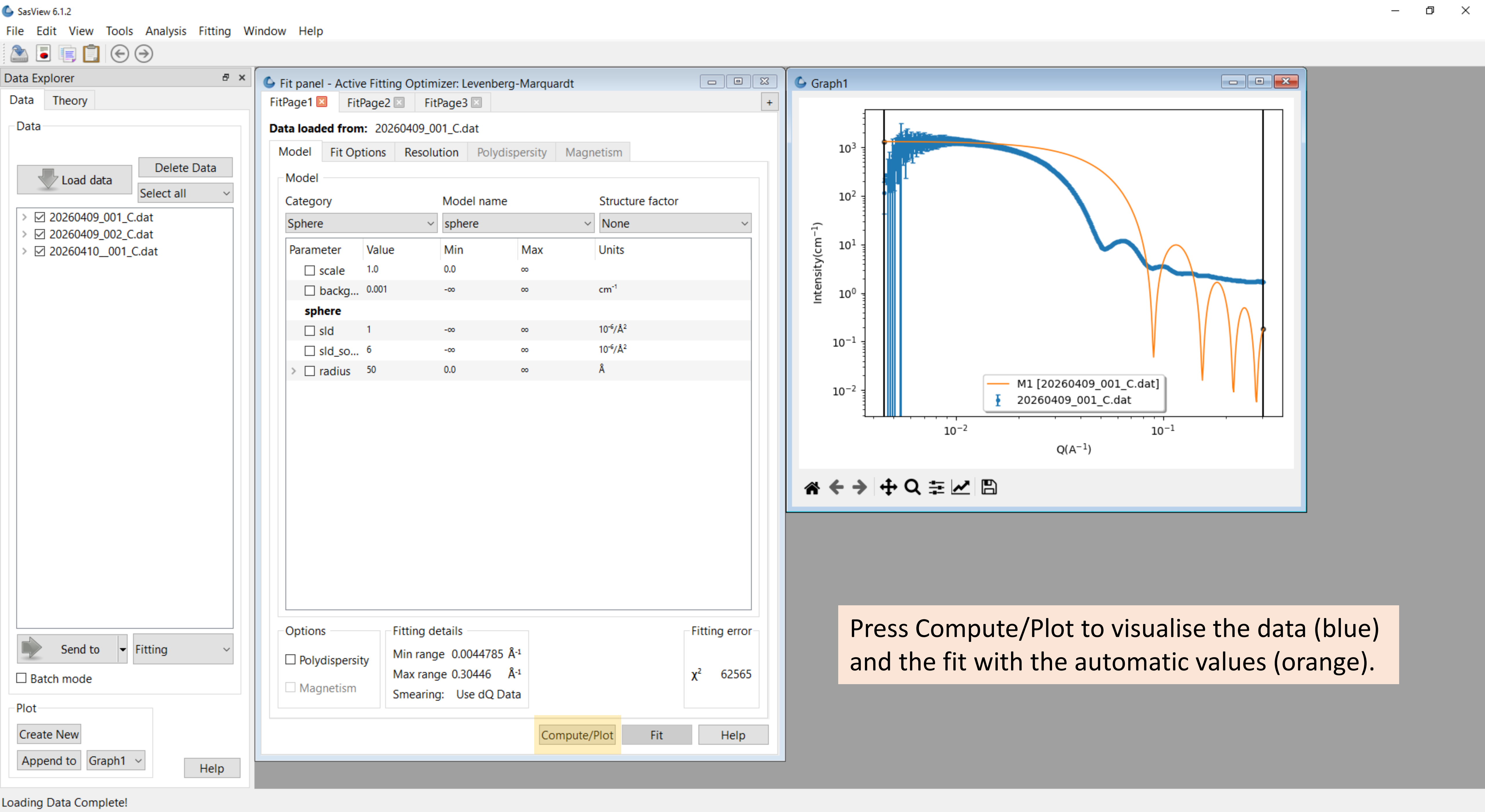
Press « load data » and select your data files generated by Igor.



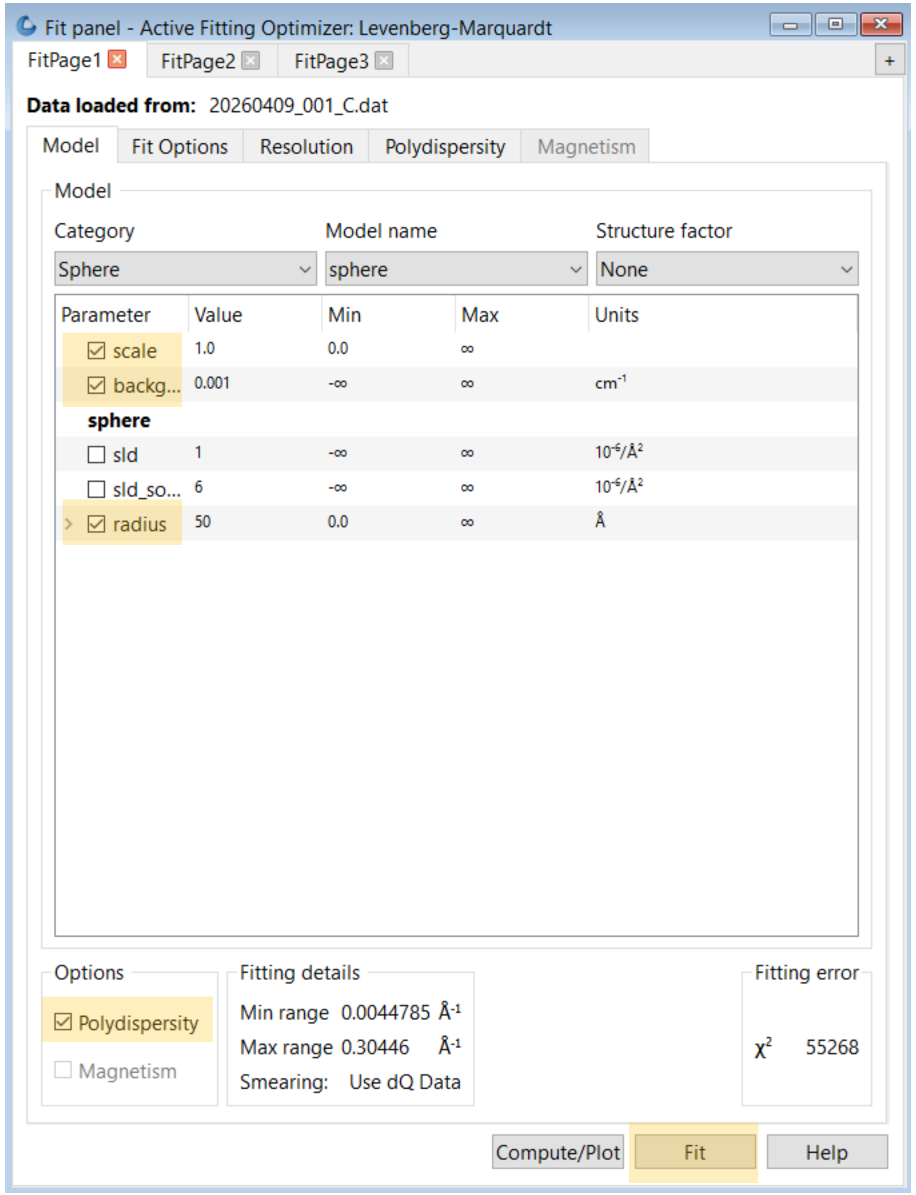
Send them to fitting.
It will open a tab per data set.

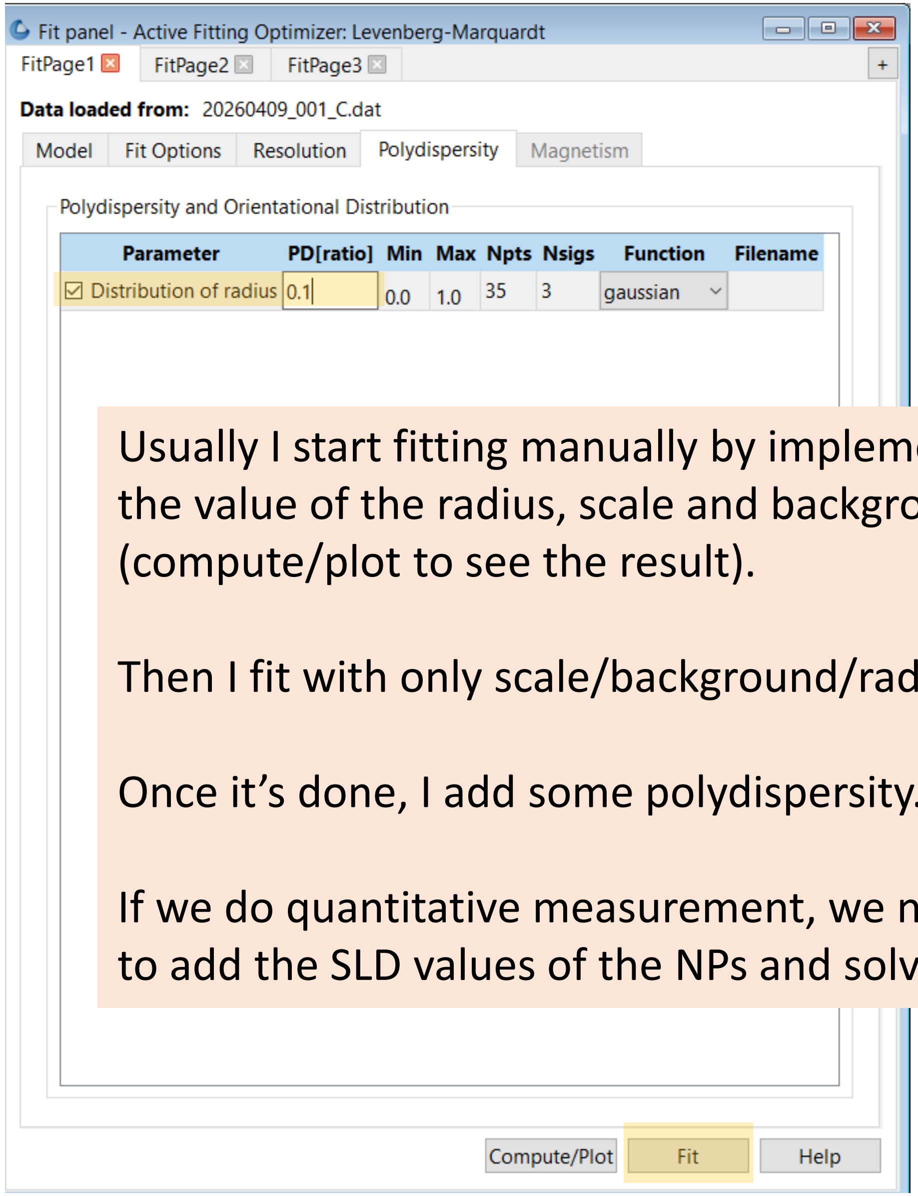


Select the model you want. Here we have simple spheres.



Press Compute/Plot to visualise the data (blue) and the fit with the automatic values (orange).



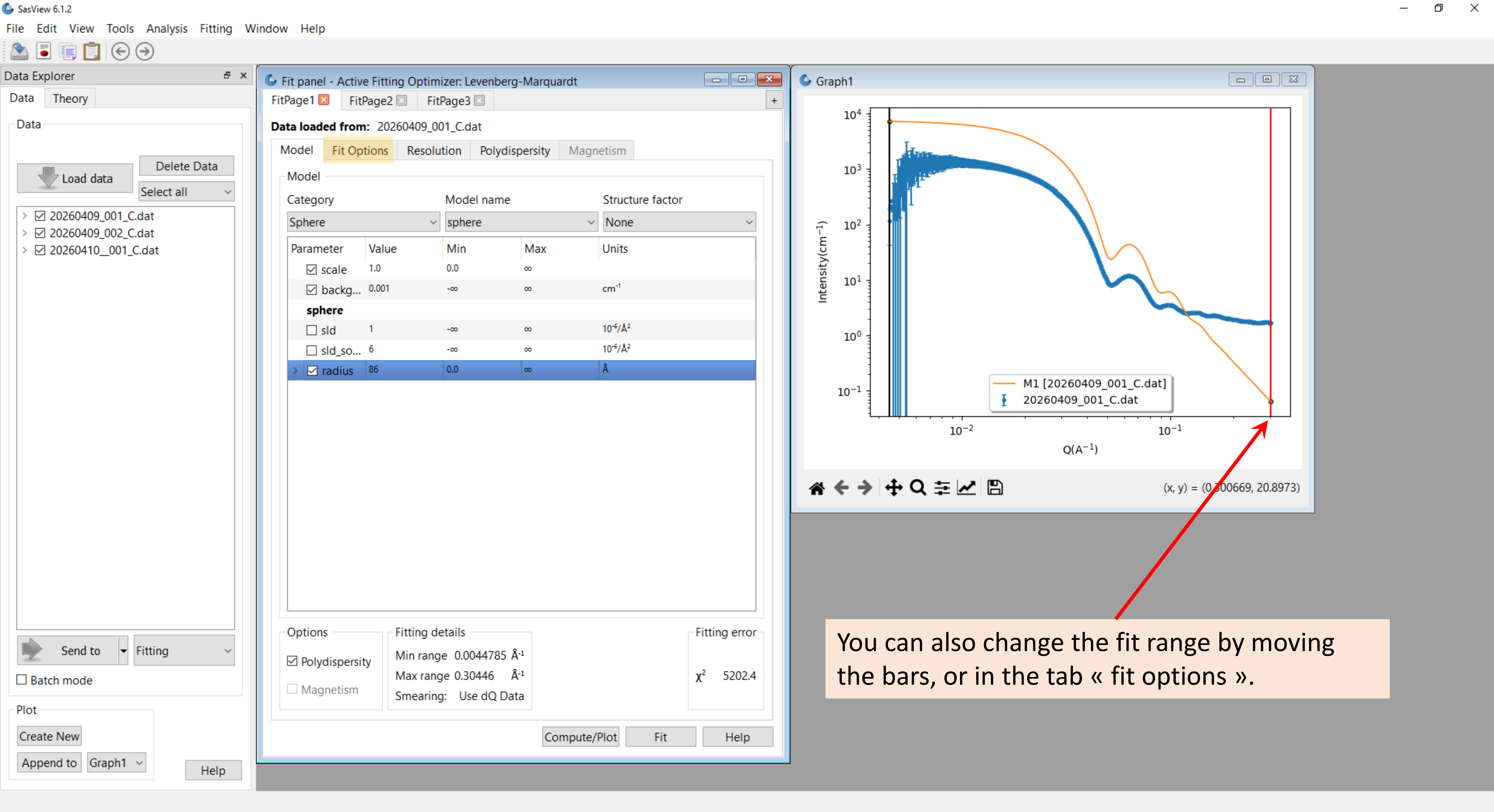


Usually I start fitting manually by implementing the value of the radius, scale and background (compute/plot to see the result).

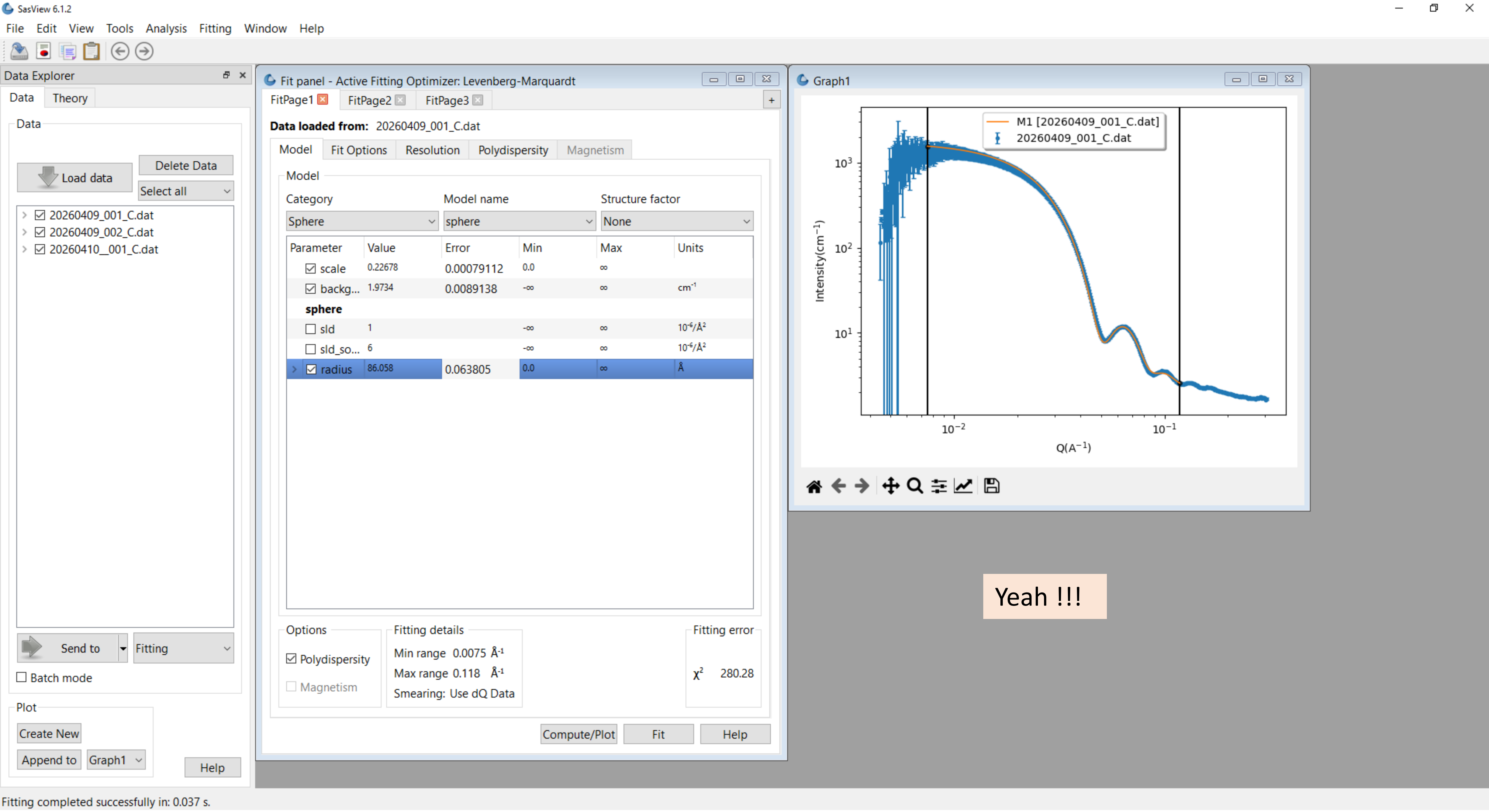
Then I fit with only scale/background/radius.

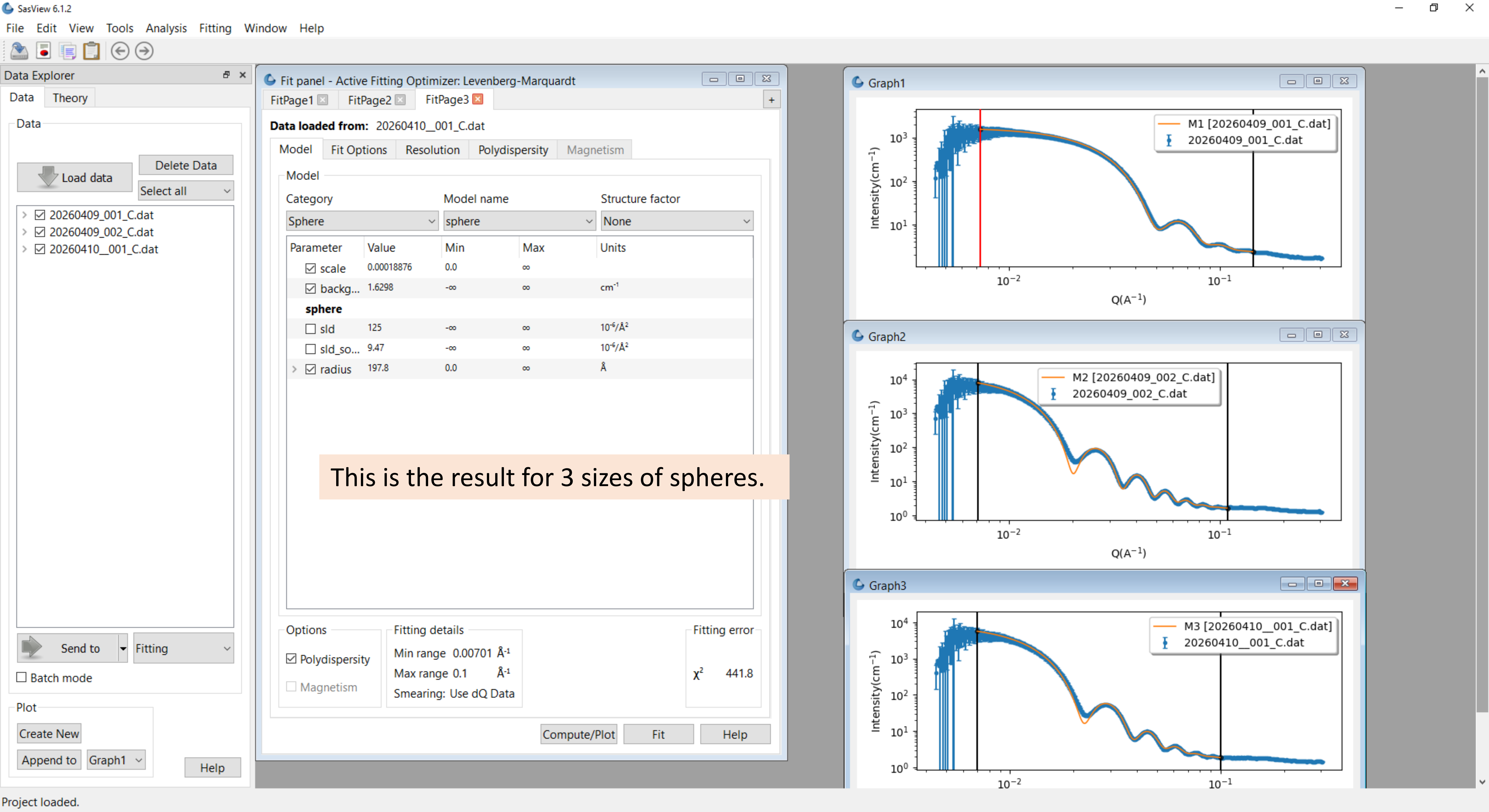
Once it's done, I add some polydispersity.

If we do quantitative measurement, we need to add the SLD values of the NPs and solvent.



You can also change the fit range by moving the bars, or in the tab « fit options ».


SasView 6.1.2
File Edit View Tools Analysis Fitting Window Help
Data Explorer
Data
Theory
Load data
Delete Data
Select all
20260409_001_C.dat
20260409_002_C.dat
20260410__001_C.dat
Send to
Fitting
Batch mode
Plot
Create New
Append to
Graph1
Help
Fit panel - Active Fitting Optimizer: Levenberg-Marquardt
FitPage1
FitPage2
FitPage3
Data loaded from: 20260409_001_C.dat
Model
Fit Options
Resolution
Polydispersity
Magnetism
Category
Model name
Structure factor
Sphere
sphere
None
Parameter Value Error Min Max Units
scale 0.22678 0.00079112 0.0 ∞
backg... 1.9734 0.0089138 -∞ ∞ cm⁻¹
sphere
sld 1 -∞ ∞ 10⁻⁶/Å²
sld_so... 6 -∞ ∞ 10⁻⁶/Å²
radius 86.058 0.063805 0.0 ∞ Å
Options
Polydispersity
Magnetism
Fitting details
Min range 0.0075 Å⁻¹
Max range 0.118 Å⁻¹
Smearing: Use dQ Data
Fitting error
χ² 280.28
Compute/Plot
Fit
Help
Graph1
M1 [20260409_001_C.dat]
20260409_001_C.dat
Intensity(cm⁻¹)
Q(A⁻¹)
Fitting completed successfully in: 0.037 s.
Yeah !!!



SasView 6.1.2
File Edit View Tools Analysis Fitting Window Help
Data Explorer
Data Theory
Load data
Delete Data
Select all
20260409_001_C.dat
20260409_002_C.dat
20260410__001_C.dat
Send to Fitting
Batch mode
Plot
Create New
Append to Graph1
Help
Fit panel - Active Fitting Optimizer: Levenberg-Marquardt
FitPage1 FitPage2 FitPage3
Data loaded from: 20260410__001_C.dat
Model Fit Options Resolution Polydispersity Magnetism
Category: Sphere
Model name: sphere
Structure factor: None
Parameter Value Min Max Units
scale 0.00018876 0.0 ∞
backg... 1.6298 -∞ ∞ cm⁻¹
sphere
sld 125 -∞ ∞ 10⁻⁶/Å²
sld_so... 9.47 -∞ ∞ 10⁻⁶/Å²
radius 197.8 0.0 ∞ Å
This is the result for 3 sizes of spheres.
Options
Polydispersity
Magnetism
Fitting details
Min range 0.00701 Å⁻¹
Max range 0.1 Å⁻¹
Smearing: Use dQ Data
Fitting error
χ² 441.8
Compute/Plot Fit Help
Graph1
M1 [20260409_001_C.dat]
20260409_001_C.dat
Graph2
M2 [20260409_002_C.dat]
20260409_002_C.dat
Graph3
M3 [20260410__001_C.dat]
20260410__001_C.dat
Intensity(cm⁻¹)
Q(A⁻¹)
Project loaded.